

Deadstock Management System

Project Report for UCS503P

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1 Higher-order goal

... *secondary framing*

This project aims to reduce capital wastage and material waste in retail supply chains by giving unsold (“deadstock”) inventory a structured path back into the market. While the broader goal is sustainable, efficient retail, the immediate engineering objective is a deployable web platform that lets vendors track, verify, and resell aging stock to other buyers.

2 Time-to-value

... *primary heuristic*

- We will deliver meaningful increments quickly by prototyping the core *list-and-browse* workflow first, before layering on verification and compliance checks.
- We will prioritize fast feedback over perfection by using weekly iterations and CI-backed automation early to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration (a working vendor-to-buyer listing loop), then improved based on user feedback.

3 Problem Statement

An estimated 20–30% of retail inventory value in India sits idle as slow-moving or dead stock annually, and over 2,00,000 licensed liquor and FMCG retailers alone face recurring dead-stock write-offs each year. Vendors currently have no structured way to recover this trapped capital. Common pain points include:

- **No real-time visibility:** vendors track stock in spreadsheets or registers, so items are only noticed *after* they’ve become deadstock.
- **No way to verify deadstock:** buyers cannot trust a listing without proof of condition, quantity, or authenticity, since none of this is independently confirmed today.
- **No resale marketplace:** even genuinely resellable stock has no dedicated channel connecting vendors with willing buyers.

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- **Compliance risk:** regulated categories (e.g., licensed liquor) cannot resell freely; any resale channel must respect licensing rules.

Why this matters now (contextual relevance): unsold stock is routinely discarded, discounted below cost, or destroyed — a direct loss to the vendor and to the environment — while a trusted resale channel could recover that value instead.

4 Proposed Solution

... Software Proposition

4.1 Overview

Build a **web-based verified marketplace** where vendors track, audit, and resell deadstock inventory to other buyers, with the following components:

- **Vendor inventory dashboard:** digital stock logging with automated aging alerts flagging items drifting toward deadstock status.
- **Verification layer:** photo/batch proof plus rule checks confirming condition, quantity, and authenticity before a listing goes live.
- **Marketplace listings:** verified deadstock published, categorized, and priced for discovery by other buyers.
- **Transaction flow:** buyers browse trusted listings and purchase directly from vendors.
- **Compliance guardrails:** category-aware rules so regulated stock (e.g., licensed liquor) cannot be listed outside permitted channels.

4.2 Core workflow

1. Vendor logs stock digitally; the system flags items aging toward deadstock status.
2. Vendor submits a deadstock item for verification (photo/batch proof).
3. System applies rule checks (completeness, category compliance) and a human/automated review confirms the listing.
4. Verified listing is published to the marketplace; buyers browse, filter, and purchase.
5. Vendor and buyer view transaction and listing status through to completion.

4.3 Operational constraints for an educational, tier-2/3 setting

- Keep the solution usable on low-to-mid range devices via responsive web design, since many vendors operate small retail outlets.
- Avoid dependence on advanced research algorithms (e.g., no computer-vision authenticity detection in scope); rely on structured proof plus rule-based checks.
- Design for deployability using common web hosting and managed databases.

5 Solution Approach

... Engineering focus

This is an engineering course project, so we focus on scalable marketplace and workflow aspects rather than novel algorithm research.

5.1 Verification assistance

... non-research

Verification will be checklist- and rule-based, not ML-driven:

- Require photo/batch proof and structured fields (category, quantity, condition) at listing time.
- Apply deterministic rule checks (e.g., category-specific compliance rules, completeness checks) before publishing.
- Flag ambiguous or regulated-category listings for manual review; never auto-publish a regulated item without a pass.

5.2 Web architecture

... web-based solution

A typical 3-tier web architecture:

- **Frontend:** responsive UI for vendor dashboard, listing creation, and buyer marketplace/browse pages.
- **Backend API:** authentication, inventory CRUD, verification rule engine, listing and transaction logic.
- **Database:** vendors, buyers, inventory items, verification records, listings, transaction history.

5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features.

6 Evaluation Criterion

... measurable and attributable

We define success using measurable metrics tied to the core workflow.

6.1 Primary evaluation metric

Time-to-listing (TTL): time between a vendor flagging an item as deadstock and it appearing as a verified, live marketplace listing.

- Target: median TTL ≤ 24 hours during pilot window.
- Attribution: measured from database timestamps (item flagged vs. listing published).

6.2 Secondary evaluation metrics

- **Recovery rate:** percentage of listed deadstock items sold within the pilot window.
- **Verification pass rate:** percentage of submitted listings passing rule checks without manual escalation.
- **Buyer trust:** buyer-reported confidence in listing accuracy using a simple 1–5 feedback question.
- **Reliability:** availability of staging/production for browsing and listing during business hours (e.g., $\geq 99\%$ uptime in pilot).

6.3 Pilot validation plan

... *high-level*

- Recruit 10–20 pilot vendors and a matching pool of buyers (simulated buyer roles if real ones are unavailable).
- Compare recovery rate and time-to-listing before/after within the iteration window, using baseline estimates from vendor interviews.

7 Scalability

... *with foresight*

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in number of vendors, listings, and concurrent buyers by:
 - decoupling components (frontend/backend),
 - enabling pagination and indexing for common marketplace queries,
 - using stateless API design.
2. **Operationally deployable (practically):** The system should run on common web hosting:
 - managed database,
 - containerized or platform-hosted deployment,
 - CI/CD pipeline for staging/production.

8 Engine availability heuristic

A mature set of “engines” (tooling/services) is available to implement this as a web-based project:

- Web framework for REST APIs and templated/reactive pages.
- Managed relational database (for structured inventory/listing/transaction data).
- Authentication approach (token-based or session-based) for vendor and buyer accounts.
- Object storage for verification photos and batch-proof documents.
- Test framework for unit/integration tests.
- CI runner and staging deployment target.

We will use these mature components to focus on marketplace design, verification correctness, and measurement rather than inventing new infrastructure.

9 Project scope and deliverables

... *iteration-friendly*

9.1 Initial deliverable

... *first iteration*

- Vendor inventory entry + aging alert logic
- Listing creation with photo/proof upload
- Rule-based verification checklist (non-regulated categories)

- Buyer-facing browse/search page for live listings
- Basic logging and timestamps for TTL measurement
- CI: build + backend unit tests + linting
- CD to staging with a single command or pipeline job

9.2 Subsequent deliverables

- Regulated-category compliance rules and manual-review escalation path
- Transaction/purchase flow between buyer and vendor
- Vendor and admin dashboards for recovery-rate metrics
- Improved search/filtering and indexed queries for scale readiness

10 Risks and mitigations

- **Regulatory risk:** restrict regulated categories (e.g., licensed liquor) from auto-publishing; require manual compliance review before those listings go live.
- **Trust/fraud risk:** require photo/batch proof at listing time and keep an audit trail of verification decisions.
- **Vendor adoption:** keep listing creation to a minimal number of steps; provide clear aging alerts so vendors act before stock is a total write-off.
- **Evaluation measurement ambiguity:** instrument timestamps and define clear listing-status transitions before development begins.

11 Summary

This project proposes a deployable web marketplace that turns unsold vendor stock into recovered revenue, by combining inventory tracking, rule-based verification, and a buyer-facing resale channel. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially time-to-listing and recovery rate). It remains focused on engineering workflow, trust, and scalability readiness rather than research-driven novelty.